Assignment #6 for Numerical Analysis

Deadline: Apr. 8

1. (10 points) Can a degree 6 polynomial intersect a degree 5 polynomial in exactly seven points? Explain.

2. (25 points) How many degree d polynomials pass through the three points $(0,1),(2,3),(3,0)$? Write one down if possible. (a) $d=1$ (b) $d=2$ (c) $d=4$.

3. (25 points) Use Lagrange interpolation to find a polynomial that passes through the points $(0,-2),(2,1),(4,4)$.

4. (40 points) Let $P_3(x)$ be the interpolating polynomial for the data $(0,0),(0.5,y),(1,3)$ and $(2,2)$. Find y if the coefficient of x^3 in $P_3(x)$ is 6.

5. (optional 40 points) The problem of interpolating m data values $\{y_j\}$ in m distinct data points $\{x_j\}$ by a polynomial of degree $\leq m-1$ was expressed in equation (1) as a square *Vandermonde* linear system of equations. Prove that this *Vandermonde* matrix is nonsingular. Thus if the interpolation problem has a solution, it must be unique.

$$\begin{bmatrix} 1 & x_1 & x_1^2 & & x_1^{m-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{m-1} \\ 1 & x_3 & x_3^2 & & x_3^{m-1} \\ & \vdots & & & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^{m-1} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{m-1} \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix} \tag{1}$$